Sequential Updates for Static Active Sensing

Problem. Recover a fixed image by sequential measurements, reusing old candidates to reduce online computation while preserving uncertainty useful for choosing the next query.

1. Setting, available prior, and target

Let $f^\star \in \mathbb{R}^d$ be the ground-truth (GT) image, d its pixel count, and f a possible complete image. At acquisition round t, choose a sensing operator $A_t \in \mathcal{A} \subset \mathbb{R}^{m\times d}$ and acquire

$$y_t = A_t f^\star + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2 I_m), \qquad H_t = ((A_1, y_1), \ldots, (A_t, y_t)). \tag{1}$$

m is the number of readings per query; $y_t \in \mathbb{R}^m$ is the actual GT measurement, shared by all K candidates. Sensor errors are independent with known variance $\sigma^2 > 0$. Generation seeds are independent of GT and sensor noise. The query policy sees only acquired readings and generated candidates, never unacquired true readings.

Measurement budget. The total number of acquisition rounds is limited to T. Given an initial history H_{t_0}, at most $T - t_0$ additional sensing operators may be selected, one per round from the admissible set $\mathcal{A}_{t-1} \subseteq \mathcal{A}$.

Prior and sampler B. $p_0(f)$ is a pretrained, frozen image prior. $B(\cdot \mid H_t)$ uses that prior and training-free measurement guidance from all of H_t to generate complete-image candidates. B is a sampling procedure, not a separately trained network.

Target. Approximate the posterior $p(f \mid H_t)$ and minimize expected reconstruction error after T queries. For $K \geq 2$ candidates $g_t^{(i)}$, $i = 1, \ldots, K$, weights $w_t^{(i)} \geq 0$ with $\sum_i w_t^{(i)} = 1$ describe their shares in the ensemble. Use their weighted average to reconstruct:

$$\mathbb{E}\left[\frac{1}{d}\left\|\sum_{i=1}^{K} w_T^{(i)} g_T^{(i)} - f^\star\right\|_2^2\right]. \tag{2}$$

Neither update route below is assumed to sample the exact posterior.

2. Sequential procedure

Step 1. Initialize candidates and weights.

At initial round t_0, draw

$$g_{t_0}^{(i)} \sim B(\cdot \mid H_{t_0}), \qquad w_{t_0}^{(i)} = 1/K, \qquad i = 1, \ldots, K.$$

Keep the candidates and their weights. Freeze model parameters at deployment.

Step 2. Select the next query and observe.

Choose the next sensing operator by weighted candidate disagreement, acquire y_t, and append (A_t, y_t) to H_{t-1}:

$$A_t \in \arg\max_{A\in\mathcal{A}_{t-1}} \sum_{i=1}^{K} w_{t-1}^{(i)} \left\| A g_{t-1}^{(i)} - \sum_{j=1}^{K} w_{t-1}^{(j)} A g_{t-1}^{(j)} \right\|_2^2. \tag{3}$$

$\mathcal{A}_{t-1}$ is the allowed query set; $Ag_{t-1}^{(i)}$ is candidate i's predicted reading, not y_t.

Step 3. Update each candidate, then update its weight.

Use one of the two candidate-update routes below, together with the same proposed recursive weighting rule.

Candidate update. For each old candidate, generate one new candidate conditioned on the latest measurement:

$$g_t^{(i)} \sim p_\theta\Big(f \mid g_{t-1}^{(i)}, A_t, y_t\Big), \qquad i = 1, \dots, K.$$

Here p_θ denotes the candidate-update distribution specified by the chosen route below, not necessarily the exact posterior.

Weight definition. We aim to assign candidate i a weight based on its posterior score under the complete measurement history:

$$w_t^{(i)} \propto p\Big(g_t^{(i)} \mid H_t\Big) = p\Big(g_t^{(i)} \mid A_1, \dots, A_t, y_1, \dots, y_t\Big), \qquad \sum_{i=1}^{K} w_t^{(i)} = 1. \tag{4}$$

Proposed recursive weight update. Multiply the previous weight by the conditional density of the new candidate given its old candidate and the new measurement; then normalize over the K candidates:

$$\widetilde{w}_t^{(i)} := p\Big(g_t^{(i)} \mid g_{t-1}^{(i)}, A_t, y_t\Big)\, w_{t-1}^{(i)}, \qquad w_t^{(i)} := \frac{\widetilde{w}_t^{(i)}}{\sum_{j=1}^{K} \widetilde{w}_t^{(j)}}. \tag{5}$$

This is a proposed finite-candidate scoring rule, not an established exact Bayesian posterior update. Routes 3.A and 3.B specify the conditional factor in this rule.

3.A. Training-free guided update [PROPOSAL]

Instead of a trained updater, freeze the prior and add measurement and old-image guidance to the base velocity $v_{\mathcal{B}}$:

$$\frac{dz^{(i)}(s)}{ds} = v_{\mathcal{B}}(z^{(i)}(s), s) - \nabla_z \left[\frac{\|y_t - A_t z\|_2^2}{2\sigma^2} + \frac{\|z - g_{t-1}^{(i)}\|_2^2}{2\tau^2} \right]_{z=z^{(i)}(s)}. \tag{6}$$

The two explicit factors are

$$p(y_t \mid A_t, g_t^{(i)}) = \mathcal{N}(y_t; A_t g_t^{(i)}, \sigma^2 I_m), \qquad p_{\text{prox}}(g_t^{(i)} \mid g_{t-1}^{(i)}) = \mathcal{N}(g_t^{(i)}; g_{t-1}^{(i)}, \tau^2 I_d). \tag{7}$$

For a fixed old candidate, the conditional factorization motivates the training-free weight update (based on Eq. (5)):

$$\begin{aligned} p(g_t \mid g_{t-1}, A_t, y_t) &\propto p(y_t \mid A_t, g_t)\, p_{\text{prox}}(g_t \mid g_{t-1}), \\ \widetilde{w}_t^{(i)} &\propto w_{t-1}^{(i)}\, p(y_t \mid A_t, g_t^{(i)})\, p_{\text{prox}}(g_t^{(i)} \mid g_{t-1}^{(i)}), \end{aligned} \tag{8}$$

so, after normalization,

$$w_t^{(i)} \propto \mathcal{N}\Big(y_t; A_t g_t^{(i)}, \sigma^2 I_m\Big)\, \mathcal{N}\Big(g_t^{(i)}; g_{t-1}^{(i)}, \tau^2 I_d\Big)\, w_{t-1}^{(i)}.$$

This is a discrete pair-scoring rule: the conditional's normalizer may depend on the old candidate, and the guidance formula does not prove that the sampler's output density equals this Gaussian product.

3.B. Sequentially trained updater: old image and new measurement as condition inputs [PROPOSAL]

$$z^{(i)}(0) \sim \mathcal{N}(0, I_d), \qquad \frac{dz^{(i)}(s)}{ds} = v_\phi\Big(z^{(i)}(s), s; g_{t-1}^{(i)}, A_t, y_t\Big), \qquad g_t^{(i)} = z^{(i)}(1). \tag{9}$$

Here $s \in [0,1]$ is internal flow time. The output law is $q_\phi(\cdot \mid g_{t-1}^{(i)}, A_t, y_t)$; q_ϕ denotes the updater's *conditional generation density*.

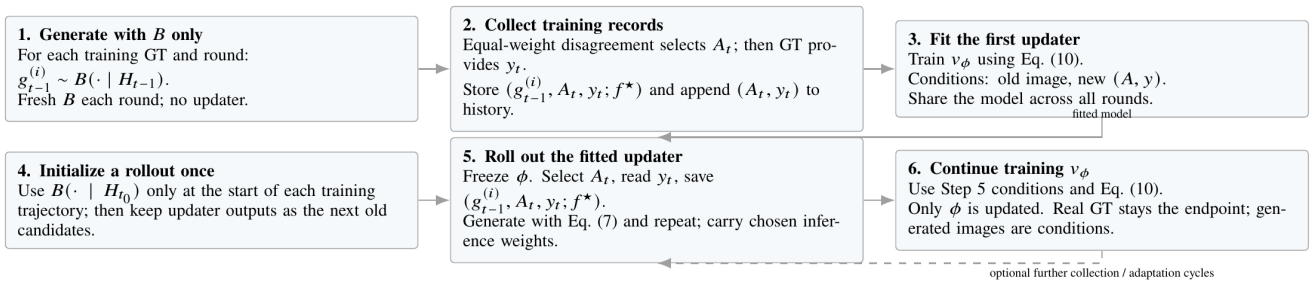
For this route, use the learned conditional generation density q_ϕ in the proposed weight update:

$$\widetilde{w}_t^{(i)} := q_\phi\Big(g_t^{(i)} \mid g_{t-1}^{(i)}, A_t, y_t\Big)\, w_{t-1}^{(i)}, \qquad w_t^{(i)} := \frac{\widetilde{w}_t^{(i)}}{\sum_{j=1}^K \widetilde{w}_t^{(j)}}. \tag{10}$$

Thus the proposal is "old weight × conditional generation density, then normalize." Retain the new candidates and weights and continue to the next acquisition round.

Sequential training for Route 3.B.

Data access. Split GT images into training, validation, and test sets. GT provides supervision and answers selected queries via Eq. (1); fixed partial logs alone cannot answer arbitrary queries. The prior stays frozen.



Training loss and inputs. Independently draw generation noise $z_0 \sim \mathcal{N}(0, I_d)$ and flow time $s \sim \mathrm{Unif}[0,1)$ for each record. Interpolate z_s and minimize

$$z_s = (1-s)z_0 + s f^\star, \qquad \mathbb{E}\Big[\big\|v_\phi(z_s, s; g_{t-1}, A_t, y_t) - (f^\star - z_0)\big\|_2^2\Big]. \tag{11}$$

Inputs are $(z_s, s, g_{t-1}, A_t, y_t)$; the output is velocity, supervised by $f^\star - z_0$, with GT as endpoint. Steps 4–6 align conditions with recursive inference, using the same network across acquisition rounds.

3. TODO / Next Steps

1. Linear case. First focus on the linear sensing setting.

- **Baselines:** ridge regression and a from-scratch ("brute-force") flow-matching model.
- **Training-free method:** first validate the proposed guidance-based candidate update and recursive weight update.
- **Sequential training if needed:** if the training-free approach is insufficient, evaluate the sequentially trained updater in Route 3.B.

2. Future work: nonlinear case. Explore nonlinear sensing settings and recurrent/state-space architectures (e.g., Mamba) for maintaining a recursive representation of the acquisition history and conditioning posterior updates.

4. Reference, previous weight attempts, and related work

FPPF reference (using our symbols). For a dynamically evolving hidden image, FPPF generates $g_t^{(i)}$ from a proposal density $q_\phi(\cdot \mid g_{t-1}^{(i)}, A_t, y_t)$ and updates

$$w_t^{(i)} \propto w_{t-1}^{(i)} \frac{p(y_t \mid A_t, g_t^{(i)})\, p(g_t^{(i)} \mid g_{t-1}^{(i)})}{q_\phi(g_t^{(i)} \mid g_{t-1}^{(i)}, A_t, y_t)}. \tag{12}$$

Here $p(y_t \mid A_t, f)$ is the sensor likelihood. The other numerator factor is the physical-state transition density. For our static true image, that transition is a Dirac point mass, not a density equal to one; computational candidate updates are distinct from physical dynamics.

Previous learned-route attempts. The initial experiment deleted the physical transition factor,

$$w_t^{(i)} \propto w_{t-1}^{(i)} \frac{p(y_t \mid A_t, g_t^{(i)})}{q_\phi(g_t^{(i)} \mid g_{t-1}^{(i)}, A_t, y_t)}; \qquad \text{alternative: } w_t^{(i)} = 1/K. \tag{13}$$

The ratio is heuristic. In preliminary trials, the team observed weight degeneracy and better performance with uniform weights. Effective sample size $\mathrm{ESS}_t = 1/\sum_i (w_t^{(i)})^2 \in [1, K]$ measures weight concentration, but ESS alone does not establish image diversity.

Related work.

AdaSense (Elata et al., ECCV 2024) selects linear queries using pretrained diffusion posterior samples.

ADS: Active Diffusion Subsampling (Nolan et al., TMLR 2025) interleaves guided diffusion with maximum-entropy query selection.

Neither requires task-specific retraining. Both are full acquisition methods; B denotes only sampling under a fixed history.

Sources: Elata et al., *Adaptive Compressed Sensing with Diffusion-Based Posterior Sampling*, 2407.08256; Lipman et al., *Flow Matching for Generative Modeling*, 2210.02747; Nagda et al., *Generative Model Proposal based Particle Filtering for Data Assimilation*, 2607.01012; Nolan et al., *Active Diffusion Subsampling*, 2406.14388; Grathwohl et al., *FFJORD: Free-form Continuous Dynamics for Scalable Reversible Generative Models*, 1810.01367.